Signal Processing Technique for Three-Level Magnetic Recording

Rakchat Charoenying, Anawin Khametong, and Chanon Warisarn

School of Integrated Innovative Technology, King Mongkut's Institute of Technology Ladkrabang, Bangkok 10520, Thailand

In the digital age, the significant growth of data has kept hard disk drives as a primary storage solution due to their cost-effectiveness and high capacity. However, the traditional perpendicular magnetic recording (PMR) technology is nearing a physical limit known as the "superparamagnetic limit," which restricts areal density (AD) to about 1 Terabit per square inch. This work introduces a simple channel model for three-level recording on granular media, along with signal processing techniques to enhance ADs. Additionally, a three-level Viterbi detector has been developed to process readback signals obtained from three-level granular media. Simulation results show that the proposed system significantly reduces the bit error rate and outperforms traditional recording systems.

***Index Terms*—Three-level magnetic recording, three-level Viterbi detector, areal density, granular media.**

I. Introduction

Currently, perpendicular magnetic recording (PMR) [1] is the dominant magnetic recording technology in practical applications. It utilizes granular media composed of small, magnetically isolated grains. Increasing the areal density (AD) of PMR systems requires reducing the bit size on the recording medium, which is determined by the bit length (BL) and track pitch (TP). As a result, the magnetic grain size must also be reduced to maintain an acceptable signal-to-noise ratio (SNR). However, excessive reduction in grain size leads to the superparamagnetic limit [2], where magnetic states lose thermal stability. Consequently, the achievable AD of PMR systems is constrained to approximately 1 terabit per square inch (Tb/in^2).

To overcome these limitations and to accommodate the increasing demand for storage capacity, several advanced magnetic recording technologies have been developed [3]-[5]. Moreover, in recent years, researchers have introduced a novel magnetic recording medium together with a three-level recording approach [6] based on granular FePt-L1_0 material. In conventional heat-assisted magnetic recording (HAMR) systems, thermal assistance is applied before the use of an external magnetic field to write only two binary states, "+1" and "−1." In contrast, the newly proposed medium enables three recording levels: "+1," "−1," and "0." The "0" state is achieved solely through thermal assistance, without an external magnetic field, significantly reducing magnetic energy within the grains and randomizing their magnetic orientations, leading to mutual magnetic cancellation. This technique can enhance the user density (UD) by up to 58.5% compared with conventional binary magnetic recording media [6].

To prepare for this emerging technology, conventional signal-processing techniques, such as partial-response maximum-likelihood (PRML) detection [7], are not directly applicable to three-level magnetic recording (TLMR). Consequently, a new signal processing framework must be developed to accommodate the TLMR system. In this research, a TLMR channel model is presented together with a newly designed PRML detection scheme capable of detecting three-level data symbols. The proposed system is evaluated in terms of bit error rate (BER) under various SNR conditions. Performance comparisons are conducted between the proposed TLMR system with the newly designed PRML detector and a conventional binary magnetic recording system employing traditional PRML detection. Experimental results demonstrate that the proposed system significantly reduces the BER and achieves superior recording performance compared with conventional binary magnetic recording systems.

II. TLMR Channel Model

Fig. 1 illustrates the simulated structure of a TLMR on granular media, which is generated using a discrete Voronoi diagram. The magnetization of each grain is determined by the position of its center of mass, based on which data bit area it falls into. For recording the data bit "+1", represented by yellow, the center of mass of the area is determined, and all grains within that area are assigned a magnetization of "+1". In the case of recording the data bit "−1", represented by blue, the same method as recording bit "+1" is applied. However, for recording the data bit "0", the same approach is used, but the magnetization of each individual grain within the specified area is uniformly randomized to either "+1" or "−1".

This causes the magnetization of the grains within that area to cancel each other out, resulting in a net magnetization close to zero, which is represented by light green [6]. The data bit writing process is under ideal writing conditions, meaning that every grain is written without any errors.

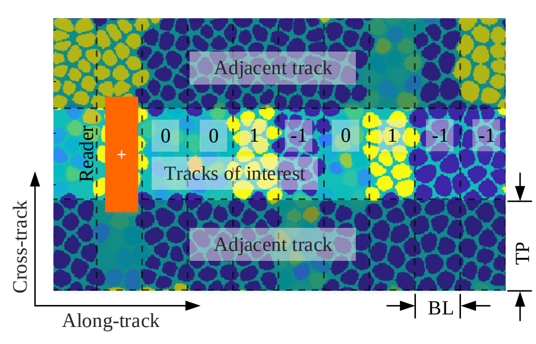


Fig. 1. Structure of a TLMR granular medium, where the yellow and blue grains represent the magnetic states "+1" and "-1", respectively, while the magnetic state "0" is represented by light green, indicating a net magnetization close to zero. The TP and BL are set to 25.25 and 12.75 nanometers, respectively, which corresponds to an AD of 2 Tb/in^2.

The readback signal, $r(t)$, obtained from the TLMR media model can be determined by the two-dimensional convolution between the media magnetization and the read head response [8], where $r(t)$ is the continuous-time readback signal. Subsequently, the readback signal is sampled at $t = kT_x$, where T_x represents the bit spacing, and k is the bit index. This results in a discrete-time readback signal, r_k, which is then used in the subsequent signal processing stages. The channel model, a simple three-level encoder/decoder, and the design of PRML signal processing can be shown in Fig. 2. During the encoding process, a sequence of binary user data bits, denoted by $u_{k'} \in \{0,1\}$, is received and subsequently encoded into a sequence of three-level data bits, denoted by $c_k \in \{0,\pm1\}$, prior to being recorded onto the media. Regarding the three-level decoding, the decoding process is performed following the three-level signal detection stage. This simple decoding is essentially the reverse operation of the encoding process. It uses the same lookup table but operates in the opposite direction, converting two three-level data bits, denoted by $\hat{c}_k \in \{0,\pm1\}$, into three binary data bits, denoted by $\hat{u}_{k'} \in \{0,1\}$, to recover the original data bits.

Fig. 3(a) illustrates the structure of a finite-state machine (FSM), a computational model that shows how data bits transition from an initial state to the next state. The data bits enclosed in circles, such as "11", "10", "01", "00", and "−1−1", represent the values currently held in memory. Each arrow indicates the next state corresponding to the incoming data bit. For the incoming three-level data bits, a solid arrow represents the bit "−1", a dotted arrow represents the bit "+1", and a dashed arrow represents the bit "0". For instance, assuming the current memory value is "00", if the incoming data bit in the channel is, the memory value after one time unit, or the next state, will transition to "10". Furthermore, the finite state machine can be represented as a trellis diagram, as shown in Fig. 3(b). The nodes on the left side represent the initial states, denoted by, while the nodes on the right side represent the next states, denoted by c_k = "+1". Each arrow symbol represents the same transitions previously described, but in this structure, they are referred to as branches. There are 9 nodes, resulting in 9×3 = 27 branches. This structure is subsequently implemented in the Viterbi detection to estimate the three-level data bits.

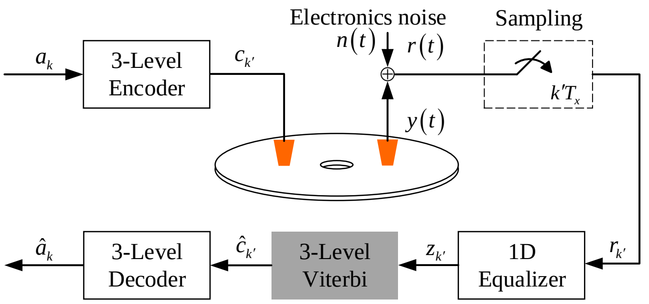
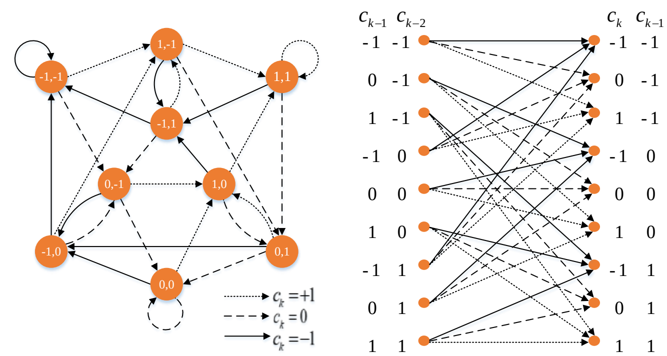


Fig. 2. Channel model of a TLMR system with the proposed PRML detection design.



(a) Finite State Machine (b) Trellis diagram

Fig. 3. Finite-state machine and Trellis diagram of the proposed three-level Viterbi detector.

III. Results and Discussion

In this study, the TLMR system features an AD of 2 Tb/in², corresponding to an UD of 3 Tb/in². In comparison, a binary magnetic recording system at an AD of 3 Tb/in² corresponds to a BL and TP of 10.50 and 21.00 nm, respectively. Thus, the BAR is set to 2 as well. We found that the proposed system significantly outperforms the conventional system. Specifically, at a BER of 10^{-4}, the proposed system demonstrates a 3 dB improvement. These results confirm that the TLMR technology, when integrated with an appropriately designed PRML detection scheme, can effectively detect three-level data. Furthermore, it shows promising potential for future developments in encoding and decoding systems to further enhance BER performance, particularly as the areal density increases beyond 3 Tb/in².

Acknowledgment

This work was financially supported by King Mongkut's Institute of Technology Ladkrabang Research Fund: Grant number KREF016820.

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